

Answers & Marking Schedule**Integration Methods Practice Problems (91579) – Merit Level****Evidence Statement**

Q	Expected Coverage	Achievement (u)	Achievement with Merit (r)
ONE (a)	$\int_{-6}^p \frac{3x + 10}{x + 7} dx = \int_{-6}^p \frac{3(x + 7) - 11}{x + 7} dx = \int_{-6}^p 3 - \frac{11}{x + 7} dx$ $= [3x - 11 \ln x + 7]_{-6}^p = 3p - 11 \ln p + 7 + 18 + 0$ $3p - 11 \ln p + 7 + 18 = 3p \quad (1)$ $\Rightarrow p = e^{\frac{18}{11}} - 7$	Reaching stage (1) of the solution.	Correct solution, with evidence of correct integration.
(b)	$\int_2^k 5e^{2x-4} dx = 10$ $\left[\frac{5}{2} e^{2x-4} \right]_2^k = 10 \quad (1)$ $\frac{5}{2} e^{2k-4} - \frac{5}{2} = 10$ $e^{2k-4} = 5$ $k = \frac{\ln(5) + 4}{2}$	Reaching stage (1) of the solution.	Correct solution, with evidence of correct integration.
(c)	$\int_1^n \left(\frac{m}{x^2} \right) dx = m - \frac{1}{2}$ $[-mx^{-1}]_1^n = m - \frac{1}{2}$ $m - \frac{m}{n} = m - \frac{1}{2} \Rightarrow n = 2m$ $\int_0^m (3x^2 - 2nx) dx = -8$ $[x^3 - nx^2]_0^m = -8$ $-m^3 = -8 \Rightarrow m = 2 \Rightarrow n = 4$	Evidence of both correct integrations.	Correct solution for both values of m and n, with evidence of correct integration.
(d)	$\int_0^k \frac{6x}{x^2 + 4} dx = \ln(125)$ $[3 \ln x^2 + 4]_0^k = \ln(125) \quad (1)$ $3 \ln(k^2 + 4) - 3 \ln(4) = \ln(125)$ $\ln\left(\frac{(k^2 + 4)^3}{64}\right) = \ln(125)$ $\frac{(k^2 + 4)^3}{64} = 125 \Rightarrow k^2 = 16$ Given that $k > 0$, $k = 4$.	Reaching stage (1) of the solution.	Correct solution, with evidence of correct integration.

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TWO (a)	$\text{Area} = \int_0^{\frac{\pi}{4}} 6 \sin(2x) \cos(2x) dx$ Recall the identity $\sin(2\theta) = 2 \sin(\theta) \cos(\theta)$ $\text{Area} = \int_0^{\frac{\pi}{4}} 3 \sin(4x) dx$ $= \left[-\frac{3}{4} \cos(4x)\right]_0^{\frac{\pi}{4}} = \frac{3}{4} - \left(-\frac{3}{4}\right) = \frac{3}{2}$	• Correct integral.	• Correct area, with evidence of correct integration.
(b)	$\text{Area} = \int_0^{\frac{\pi}{3}} A \sin(3x) \sin(2x) dx$ Recall the identity $2 \sin A \sin B = \cos(A - B) - \cos(A + B)$ $\int_0^{\frac{\pi}{3}} \frac{A}{2} (\cos(x) - \cos(5x)) dx = 6\sqrt{3}$ $\frac{A}{2} \left[\sin(x) - \frac{1}{5} \sin(5x)\right]_0^{\frac{\pi}{3}} = 6\sqrt{3}$ $\frac{A}{2} \left(\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{10}\right) = 6\sqrt{3}$ $A = 20$	• Correct integral.	• Correct solution, with evidence of correct integration.
(c)	$\text{Area} = \int_0^{\frac{\pi}{6}} \frac{2 \tan x}{1 - \tan^2 x} dx$ Recall the identity $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$ and $\tan x = \frac{\sin x}{\cos x}$ $\text{Area} = \int_0^{\frac{\pi}{6}} \tan(2x) dx = \int_0^{\frac{\pi}{6}} \frac{\sin(2x)}{\cos(2x)} dx$ $= \left[-\frac{1}{2} \ln \cos(2x) \right]_0^{\frac{\pi}{6}} = \frac{1}{2} \ln(2) - 0 = \frac{1}{2} \ln(2)$	• Correct integral.	• Correct area, with evidence of correct integration.
(d)	$e^x = 4 - 3e^{-x}$ $\Rightarrow (e^x - 1)(e^x - 3) = 0 \Rightarrow x = 0 \text{ or } x = \ln(3)$ $\text{Area} = \int_0^{\ln(3)} ((4 - 3e^{-x}) - e^x) dx$ $= [4x + 3e^{-x} - e^x]_0^{\ln(3)}$ $= \left(4 \ln(3) + 3 \times \frac{1}{3} - 3\right) - (0 + 3 - 1)$ $= 4 \ln(3) - 4$	• Correct integral.	• Correct area, with evidence of correct integration.

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THREE (a)	$v(t) = \int 1.5e^{0.5t} dt = 3e^{0.5t} + c$ $2 = 3e^{0.5 \times 0} + c \Rightarrow c = -1$ $s = \int_2^3 (3e^{0.5t} - 1) dt = [6e^{0.5t} - t]_2^3$ $= 6e^{1.5} - 6e - 1 \approx 9.58 \text{ m}$	Correct formula for velocity, including evaluating the c value.	Correct distance found, with evidence of integration methods.
(b)	$v(t) = \int 3 \cos(2t) dt = \frac{3}{2} \sin(2t) + c$ $1 = \frac{3}{2} \sin(0) + c \Rightarrow c = 1$ $s(t) = \int \left(\frac{3}{2} \sin(2t) + 1 \right) dt = -\frac{3}{4} \cos(2t) + t + d$ $2 = -\frac{3}{4} \cos(0) + 0 + d \Rightarrow d = 2.75$ $s(t) = -\frac{3}{4} \cos(2t) + t + 2.75$ $s(\pi) = -\frac{3}{4} \cos(2\pi) + \pi + 2.75 = 2 + \pi \approx 5.14 \text{ m}$	Correct formula for velocity, including evaluating the c value.	Correct distance found, with evidence of integration methods.
(c)	$v(t) = \int \frac{6e^{3t}}{(e^{3t} + 2)^2} dt = \int 6e^{3t} (e^{3t} + 2)^{-2} dt$ $v(t) = \frac{-2}{e^{3t} + 2} + c$ $2 = \frac{-2}{e^0 + 2} + c \Rightarrow c = \frac{8}{3}$ $v(\ln(2)) = \frac{-2}{e^{3 \ln(2)} + 2} + \frac{8}{3} = \frac{37}{15} \text{ m s}^{-1}$	Correct formula for velocity.	Correct exact velocity found, with evidence of integration methods.

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FOUR (a)	$y^2 dy = 4e^{2x} dx$ $\int y^2 dy = \int 4e^{2x} dx$ $\frac{y^3}{3} = \frac{4}{2} e^{2x} + c$ $\frac{3^3}{3} = 2e^0 + c \Rightarrow c = 7$ $y^3 = 6e^{2x} + 21$ $\text{When } x = \ln(2), y = \sqrt[3]{45}$	Evidence of both correct integrals.	Correct evaluation of the y value, with evidence of integration methods.
(b)	$\frac{1}{y^2} dy = \sin(3x) dx$ $\int y^{-2} dy = \int \sin(3x) dx$ $-\frac{1}{y} = -\frac{1}{3} \cos(3x) + c$ $-1 = -\frac{1}{3} \times (-1) + c \Rightarrow c = -\frac{4}{3}$ $\frac{1}{y} = \frac{\cos(3x) + 4}{3}$ $\text{When } x = \frac{\pi}{6}, y = \frac{3}{4}$	Evidence of both correct integrals.	Correct evaluation of the y value, with evidence of integration methods.

(c)	Recall the identity $2 \sin A \cos B = \sin(A + B) + \sin(A - B)$ $\frac{dy}{dx} = 8 (\sin(8x) + \sin(-2x))$ $y = \int 8 \sin(8x) - 8 \sin(2x) \, dx$ $y = -\cos(8x) + 4 \cos(2x) + c$ $5 = -\cos(4\pi) + 4 \cos(\pi) + c$ $c = 10$ $y = -\cos(8x) + 4 \cos(2x) + 10$ When $x = \frac{\pi}{4}$, $y = 9$	• Correct integral.	• Correct evaluation of the y value, with evidence of integration methods
(d)	$y \, dy = x(x^2 + 16)^{-\frac{1}{2}} \, dx$ $\int y \, dy = \int x(x^2 + 16)^{-\frac{1}{2}} \, dx$ $\frac{y^2}{2} = (x^2 + 16)^{\frac{1}{2}} + c$ $\frac{4^2}{2} = (3^2 + 16)^{\frac{1}{2}} + c \Rightarrow c = 3$ $\frac{y^2}{2} = \sqrt{x^2 + 16} + 3$ When $x = 0$, $y = \sqrt{14}$ (because $y > 0$)	• Evidence of both correct integrals.	• Correct evaluation of the y value, with evidence of integration methods

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FIVE (a)	$\frac{1 - \tan^2(x)}{\sec^2(x)} = \frac{1 - \frac{\sin^2(x)}{\cos^2(x)}}{\frac{1}{\cos^2(x)}} = \cos^2(x) - \sin^2(x) = \cos(2x)$ $\int_0^{\frac{\pi}{8}} \cos(2x) \, dx = \left[\frac{1}{2} \sin(2x) \right]_0^{\frac{\pi}{8}} = \frac{1}{2} \sin\left(\frac{\pi}{4}\right) - \frac{1}{2} \sin(0) = \frac{\sqrt{2}}{4}$	• Correct integral, but with one minor error.	• Correct solution, with evidence of correct integration.
(b)	$\frac{du}{dx} = e^x \Rightarrow dx = \frac{du}{e^x}$ $\int_2^4 \frac{e^{2x}}{u} \frac{du}{e^x} = \int_2^4 \frac{e^x}{u} \, du = \int_2^4 \frac{u-1}{u} \, du = \int_2^4 \left(1 - \frac{1}{u}\right) \, du$ $= [u - \ln u]_2^4 = (4 - \ln(4)) - (2 - \ln(2)) = 2 - \ln(2)$	• Correct integral, but with one minor error.	• Correct solution, with evidence of correct integration.